



0.3M Global Shutter MIPI Camera Module SC035HGS

Key Properties

SC035HGS Image Sensor

Output Formats: 10-bit RAW MONO/RGB

Frame Rate: 2-lane@MIPI

180fps for 640(H)x480(V)

S-mount M7 Lens Holder

Application

Portable Media / Robot Vision Camera/ Image Recognition

Laser Sensing / DIY Camera Solutions / Action Photography

Feature

- ♦ 0.3M(640x480) image sensor SC035HGS
 - Low illumination and high dynamic range sensor
- ♦ Megapixel M7 lenses optional
- ♦ 10-bit RAW MONO / RAW RGB
- ♦ MIPI CSI2 Interface

Introduction

MCM03M180M7 is a 0.3-megapixel global shutter CMOS image sensor developed by SmartSens.

Featuring a 1/6" optical format and 3.744 μm pixel pitch, it delivers high-speed imaging at up to 180 frames per second. With its backside-illuminated (BSI) pixel structure and high sensitivity (6500 mV/lux-sec), the MCM03M180M7 excels in low-light and high-dynamic-range environments. It supports either monochrome or RGB output with 10-bit ADC options and is ideal for applications such as machine vision, barcode scanning, and industrial inspection.

MCM03M180M7 uses 2-lanes MIPI CSI2 interface makes it easy to use. It has S-mount (M7) lens holder for enabling the users to choose the lens and any spectral filter as per their requirements.



**Specifications**

Model	MCM03M180M7
Sensor	SC035HGS
Image Size	1/6 inch
Pixel Size	3.744 μm
Image Area	2396 μm x 1797 μm
Image Pixels & Resolution	640(H) x 480(V)
Output Formats	10-bit 2 lane MIPI RAW MONO/RGB
Image Transfer Rate	Full resolution (640x480): 180 fps 640x360: 180 fps 320x240: 180fps
Max S/N Ratio	40dB
Dynamic Range	HDR Mode 100dB / Linear Mode 60dB
Sensitivity	Mono 6500 mV/lux-sec
Shutter Type	Global shutter
Lens Parameters	Standard: FOV(D)60°
Lens Holder	Standard M7 lens holder
Power Supply	External Input
Operating Voltage	DC 1.5V / 1.8V / 2.8V
Operating Current	85mA
Operating Temperature	-30 ~ 70°C
Stable Image Temperature	-20 ~ 60°C
Total length	50mm
FFC Pin Numbers	24
FFC Pin Pitch	0.5mm

Specifications are subject to change without notice.

Sensor Board Drawings

PIN	NAME
1	NC
2	TRIG
3	PWDN
4	NC
5	SDA
6	SCL
7	GND
8	GND
9	MIPI_RDN0
10	MIPI_RDP0
11	MIPI_RDN1
12	MIPI_RDP1
13	NC
14	NC
15	NC
16	NC
17	GND
18	MIPI_RCN
19	MIPI_RCP
20	GND
21	MCLK
22	DOVDD1.8V
24	DVDD 1.5V
24	AVDD 2.8V

Steel plate
t=0.2

Dimensions and Tolerances:

- Top View: 8±0.10, 8±0.10, 6±0.10, 12.5±0.10, 4.5±0.10
- Side View: 50±0.10, 1.7±0.10, 8±0.10, 7.00±0.3±0.00
- Front View: 4.49±0.10, 0.1±0.10, 12.5±0.10

UNLESS OTHERWISE SPECIFIED:
 DIMENSIONS ARE IN mm
 SURFACE FINISH ±0.3 OR BETTER
 DEDURROBCKAL, SHARP EDGES
 TOLERANCES: ANGLES
 MAX 0.5 - X.XX ±0.05 ±0.20
 Material: -
 Surface Finish: -

REV.	ZONE	Description	By	Date

Drawn By: Leo Chen	Title: CAMERA MODULE
Checked By: -	Prt NO: MC1003A-00M7
Approved By: D 300 HU	SCALE: 2:1
Projection Type: 1st Ang	Unit: mm
	REV: A0
	Date: 2025/8/27

Quantity: 1
 SHEET: 1 OF 1